

PYTHON. DATA ANALYSIS ESSENTIALS. Dataset creation.

Instructor: Elena Fernández Fernández

elena.fernandezfernandez@uzh.ch

The Course “**Python. Data Analysis Essentials**” ([/ Python: Data Analysis Essentials - Universität Zürich](#)) uses 19th century English Literature data. We are using **two British authors** (Elizabeth Gaskell and Charles Dickens) and **two USA authors** (Mark Twain and Edith Wharton).

The goal of the course is to try to analyze if we can detect some **patterns in terms of gender** (male/female writers) **and country of origin** (United Kingdom/United States)ⁱ.

You can find all the materials that are used for the **creation of the data outputs** that we are using as examples in this course in this repository: [GitHub - effernan/Python-Data-Analytics-Essentials-Dataset-Creation: This repository contains all the materials used for creating the Dataset of the UZH course "Python: Data Analytics"](#).

The **final metadata analysis** that we are using for our class can be found in here: [Python-Data-Analytics-Essentials-Dataset-Creation/Final Metadata Analysis/Final Metadata.ipynb at main · effernan/Python-Data-Analytics-Essentials-Dataset-Creation · GitHub](#)

The goal is to try to see if we can find any interesting patterns in terms of:

- **Temporal distribution of literary production:** [Python-Data-Analytics-Essentials-Dataset-Creation/Charles Dickens/Book titles and dates at main · effernan/Python-Data-Analytics-Essentials-Dataset-Creation · GitHub](#)
- **Length of books (i.e. counting total and unique number of words):** [Python-Data-Analytics-Essentials-Dataset-Creation/Charles Dickens/Counting Words/Word Count.ipynb at main · effernan/Python-Data-Analytics-Essentials-Dataset-Creation · GitHub](#)
- **Number of books written per author:** [Python-Data-Analytics-Essentials-Dataset-Creation/Charles Dickens/Dataset/Dataset Creation.ipynb at main · effernan/Python-Data-Analytics-Essentials-Dataset-Creation · GitHub](#)
- **Entities (Name Entity Recognition):**
 - **Geopolitical Entites (GPE):** [Python-Data-Analytics-Essentials-Dataset-Creation/Charles Dickens/Entities/GPE/Extracting GPE Entities in All Books.ipynb at main · effernan/Python-Data-Analytics-Essentials-Dataset-Creation · GitHub](#)
 - **Person Entities:** [Python-Data-Analytics-Essentials-Dataset-Creation/Charles Dickens/Entities/Person/Extracting Person Entities in All Books.ipynb at main · effernan/Python-Data-Analytics-Essentials-Dataset-Creation · GitHub](#)

- **Male and Female Characters:** [Python-Data-Analytics-Essentials-Dataset-Creation/Charles Dickens/Entities/Male and Female Characters/Counting number of male vs female characters.ipynb at main · effernan/Python-Data-Analytics-Essentials-Dataset-Creation · GitHub](#)
- **Geolocation** (included in the general metadata analysis repository).

All those links will take you to a data analysis done using Charles Dickens. If you would like to reproduce this code with our own author you just need to use your own dataset (but the scripts will work!).

ⁱ The scripts used come from **Book NLP** ([GitHub - booknlp/booknlp: BookNLP, a natural language processing pipeline for books](#)).